

ADDITIONAL PROJECT 06

GSM-Connected Gas and Flame Alarm

Arduino Safety Monitoring with Local Mitigation and SMS Notification

CONTEXT	ROLE	APPLICATION	PERIOD
Team Engineering Project	Project Team Member	Domestic Safety Monitoring	June 2022

Project Overview

Developed a breadboard safety prototype that monitors combustible-gas concentration and flame presence, presents the live gas value locally, and initiates layered responses when a hazard is detected. The system combines immediate on-site warning and ventilation with remote SMS notification so that its protective functions do not depend solely on network availability.

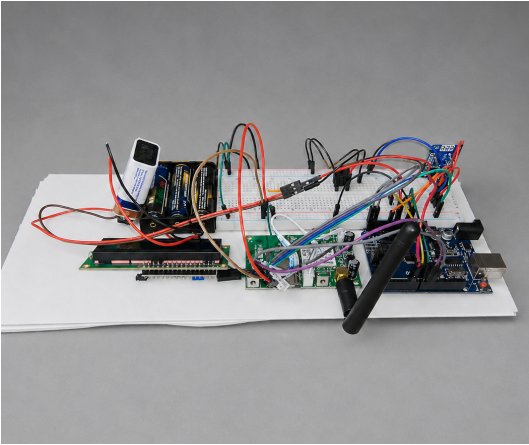
Engineering implementation

- Integrated an Arduino Uno with an analog MQ-5 gas sensor and a digital IR flame-detection module.
- Programmed continuous gas acquisition, threshold evaluation, and a 16×2 I²C LCD status interface.
- Implemented local alarm signalling through a buzzer and red/green status LEDs.
- Switched a 12 V ventilation fan through a relay while keeping the Arduino and GSM electronics on a separate 5 V supply.
- Interfaced a Neoway M660 GSM/GPRS module over UART and transmitted an SMS when the alarm condition was reached.

Deterministic response path

MQ-5 / FLAME SENSOR → ARDUINO DECISION LOGIC → LOCAL ALARM + FAN

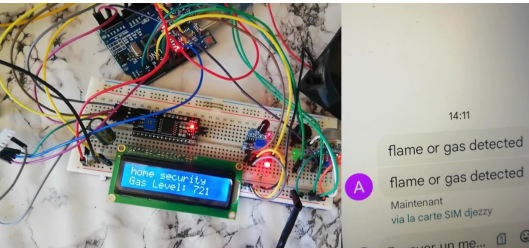
In parallel: GSM UART command sequence → remote SMS alert.



Assembled sensing, display, power, and GSM electronics. Background and lighting cleaned for presentation.

VERIFIED SYSTEM OUTCOME

A detected hazard triggered local indication, forced ventilation, and a received SMS alert.



Video evidence: displayed gas value of 721 and the received GSM alarm message.

PHYSICAL PROTOTYPE VALIDATION

The LCD displayed stable ambient readings of approximately 15–26 raw ADC counts. Controlled exposure to lighter gas increased the displayed value to 721, after which the alarm response and fan were demonstrated and the phone received the message “flame or gas detected.” These values were not shown as calibrated ppm measurements, and the video does not isolate a separate flame-only validation sequence.

PROPOSED 2026 ARCHITECTURE EVOLUTION — NOT YET IMPLEMENTED

- Add calibrated, target-selective gas sensing, a dedicated smoke detector where required, environmental compensation, and automatic sensor-fault diagnostics.
- Replace the legacy 2G modem with an LTE-M/NB-IoT module and authenticated MQTT/TLS telemetry, retaining SMS as a fallback channel.
- Preserve alarm, buzzer, and fan control as deterministic MCU functions; add a watchdog, battery backup, self-test, and network-health supervision.

TECHNOLOGIES

Arduino Uno Embedded C/C++ MQ-5 IR Flame Sensor Neoway M660 GSM / SMS UART I²C LCD Relay + 12 V Fan